

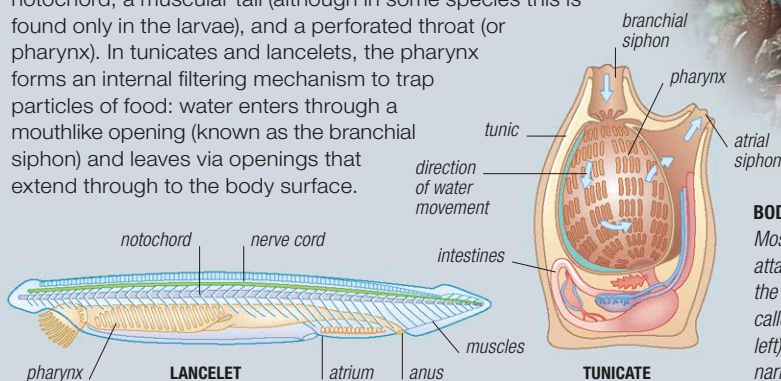
Invertebrate chordates

PHYLUM	Chordata
SUB PHyla	Urochordata Cephalochordata
CLASSES	5
ORDERS	9
FAMILIES	38
SPECIES	About 3,000

Chordates are animals that have an internal skeletal rod, called a notochord. Most chordates are vertebrates, in which the notochord is largely replaced by interlocking vertebrae. However, there 2 groups of chordates—the tunicates and lancelets—that have no vertebral column and are therefore classified as invertebrates. Both the tunicates and lancelets are marine animals. Most tunicates have a bag-shaped body, while lancelets have a long, narrow, rigid body.

Anatomy

Adult lancelets and lampreys retain their notochord, while in tunicates, it is present only in the larval stage. All species (apart from some tunicates) also have a hollow nerve cord above the notochord, a muscular tail (although in some species this is found only in the larvae), and a perforated throat (or pharynx). In tunicates and lancelets, the pharynx forms an internal filtering mechanism to trap particles of food: water enters through a mouthlike opening (known as the branchial siphon) and leaves via openings that extend through to the body surface.



BODY SECTION

Most tunicates (left) are filter feeders that live attached to the seabed. Water enters through the mouth and leaves through an opening called the atrium. The body of a lancelet (far left) is specialized for swimming: it is long, narrow, flexible, and muscular.

TUNICATE COLONIES

Colonies of tunicates often form a brightly colored mass on the ocean floor. Although the individual members of the colony have separate mouths, they sometimes share an opening for expelling water and waste products.

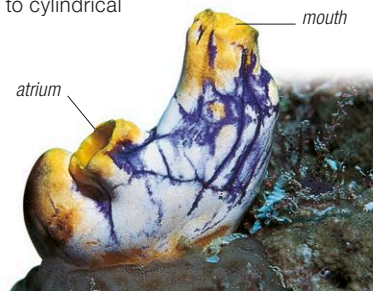
Class Ascidiacea

Sea squirts

Occurrence 2,800 spp. worldwide; attached to coastal rocks and the seabed



Sea squirts are baglike tunicates that live fastened to coastal rocks or to the seabed. Some species are predators, but most feed by pumping water through a mesh-lined pharynx, which filters out particles of food. The water is drawn in through one tube, the inhalant siphon or mouth, and expelled through another, the exhalant siphon or atrium. The body is surrounded by a large, thick, spherical to cylindrical



POLYCARPA AURATA

This solitary sea squirt from the Indo-Pacific is covered by a thick, leathery, sometimes brightly colored, protective test or tunic, 3/4–6 in (2–15 cm) high. Two siphons project from it, and these are the entry and exit sites of the water current from which the animal filters its food.



BOTRYLLUS SCHLOSSERI

In some sea squirt colonies, such as the one above, the zooids are arranged so that each has its own site of water intake, but groups share a communal water exit. In this widespread species, the groups are arranged in star patterns, and the color of the zooids—which are just 1/16 in (2 mm) across—contrasts with the colony's jellylike matrix.

tunic, or test, of cellulose-like material. Sea squirts are sometimes solitary, but many species form colonies, in which individual animals (zooids) are arranged in groups and the exhalant siphon is shared. The zooids range from 1/32–23 1/2 in (0.1) to 60 cm in height, while colonies may reach several yards high or across. Sea squirts are often of particular interest because of their affinities with vertebrate chordates. As adults, the pharynx is their only visible chordate feature, but sea squirt larvae (which look like tadpoles) have a tail that contains a notochord and a hollow, dorsal nerve cord—both vertebrate characteristics.

Class Thaliacea

Pelagic tunicates

Occurrence 78 spp. worldwide; planktonic in ocean



This class of tunicates contains free-living organisms, drifting among the plankton in the open sea, that have inhalant and exhalant openings at opposite ends of their test. Like sea squirts, pelagic tunicates have a large, perforated pharynx, which occupies most of the body mass. There are 2 basic kinds: pyrosomes, which are colonial, and salps and doliolids, which have alternating phases of solitary and colonial existence. Individuals in the colony (zooids) are up to 10 in (25 cm) long, but some complete colonies may exceed 46 ft (14 m) in length.



PYROSOMA ATLANTICUM

This colonial, luminescent, warm-water pyrosome, measuring more than 3 1/4 ft (1 m) in length, resembles a giant planktonic test tube. It filters food from the water, and uses its watery "exhaust" to propel itself slowly through the sea.

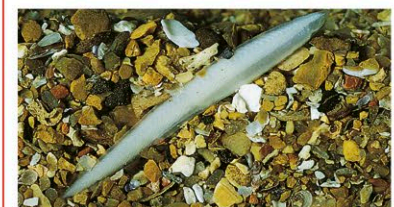
Class Leptocardia

Lancelets

Occurrence 30 spp. in tropical and temperate zones; in shallow sand or gravel



Lancelets are small—usually around 2 in (5 cm) long—semi-transparent animals with several vertebrate-like features. As well as a notochord, they have gill-like pharyngeal slits, a single dorsal fin, and—like many fish—muscle blocks arranged in V-shaped patterns down the body. Lancelets also have small tentacles around the mouth. They suck in water and food particles by beating microscopic hairs (cilia) that line their pharyngeal slits. The sexes are separate, and they reproduce by external fertilization. In some places, they are caught for food.



BRANCHIOSTOMA LANCEOLATUM

This European species grows up to 2 in (5 cm) long. Like all lancelets, it swims by wiggling its body from side to side. In its normal feeding position, most of its body is buried, with just its head projecting out of the seabed.